

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2011

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10
 - Use a different booklet for each question. Write the problem number and your **code number** (not your name) on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
-

- Let G be a finite group of order n . There is a homomorphism $G \rightarrow S_n$, where $g \in G$ maps to the permutation π_g , where π_g permutes the n elements of G by $\pi_g(x) = gx$.
 - Show that π_g is an odd permutation if and only if g has even order and $[G : \langle g \rangle]$ is odd
 - Show that if a Sylow 2-subgroup of G is non-trivial and cyclic then G has a subgroup H with $[G : H] = 2$
- Let $n \geq 1$ and let B be an $n \times n$ complex matrix. Let M_n be the set of $n \times n$ complex matrices, regarded as a vector space of dimension n^2 over $\mathbb{C}$. Let

$$\phi_B : M_n \longrightarrow M_n, \quad A \longmapsto AB.$$

Then ϕ_B is a linear transformation of M_n .

- Show that if $m = kn$ for some k with $0 \leq k \leq n$, then there is a matrix B such that $\text{Ker}(\phi_B)$ has dimension kn .
 - Let B be arbitrary. Show that the dimension of $\text{Ker}(\phi_B)$ is a multiple of n .
- Let F be a field and suppose that the additive group of F is a finitely generated abelian group. Show that F is a finite field.
 - Let R be a commutative ring with 1.
 - Suppose that R contains a non-finitely generated ideal. Show that the set of non-finitely generated ideals contains an ideal I that is maximal for that set. That is, I is not finitely generated, and if J is an ideal with $I \subsetneq J$, then J is finitely generated. (Note: You may use Zorn's Lemma)
 - Let I be as in Part (a). Suppose that there exist $x, y \in R$ with $xy \in I$ but $x, y \notin I$. Let $J = xR + I$. Show that there exist elements $g_1, \dots, g_n \in I$ (not just in R) such that $x, g_1, \dots, g_n$ generate J .
 - Let x, I be as in Part (b). Let $M = \{r \in R \mid rx \in I\}$. Show that there are $h_1, \dots, h_m \in M$ that generate the ideal M .
 - Let the notation be as in Parts (a), (b), (c). Show that $g_1, \dots, g_n, xh_1, \dots, xh_m$ generate I . (This contradicts the choice of I and we conclude that I is prime.)
 - Show that if every prime ideal of R is finitely generated, then R is Noetherian.

5. Let X_1, X_2, X_3, X_4 be variables. Let S_4 act on $L = \mathbb{Q}(X_1, X_2, X_3, X_4)$ by permuting the variables. The fixed field is $K = \mathbb{Q}(\sigma_1, \sigma_2, \sigma_3, \sigma_4)$, where

$$\sigma_1 = X_1 + X_2 + X_3 + X_4, \quad \sigma_2 = X_1X_2 + X_1X_3 + X_1X_4 + X_2X_3 + X_2X_4 + X_3X_4,$$

$$\sigma_3 = X_1X_2X_3 + X_1X_2X_4 + X_1X_3X_4 + X_2X_3X_4, \quad \sigma_4 = X_1X_2X_3X_4.$$

Let

$$\gamma = X_1X_2 + X_3X_4.$$

- (a) Show that $\text{Gal}(L/K(\gamma))$ is the dihedral group with 8 elements
 (b) Let F be the smallest Galois extension of K containing $K(\gamma)$. Show that $[F : K] = 6$
6. Let G be a finite group and let H be a normal subgroup of G with $[G : H] = n$. Let V be the n -dimensional vector space with basis $\{b_{g_1H}, \dots, b_{g_nH}\}$ corresponding to the cosets of H . Define a representation

$$\rho : G \longrightarrow GL(V),$$

where $\rho(g) : V \rightarrow V$ is defined by $\rho(g)(b_{g_iH}) = b_{gg_iH}$. Show that ρ is irreducible if and only if $G = H$.